WEEK 1 TECHNICAL PROGRESS REPORT

Project Setup and Data Source Identification
Course: COM-711 (Minor Project) | Academic Session 2026-2027

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1. Executive Summary & Objective

The primary objective of Week 1 was to establish the software engineering repository, configure Git version control, execute an exhaustive reproducible audit of the supplied property dataset, investigate access methods and terms of service across all seven public data sources specified in the project synopsis, and formalize the canonical 35-variable data dictionary.

During this week, the engineering team set up the repository architecture with clear raw, interim, and processed data boundaries; 1,200 property records across 36 canonical Jammu localities; probed real-world web endpoints.

2. Original Assigned Group Tasks

• Task 1: Set up Git repository and project structure.

• Task 2: Confirm access methods for 99acres, JDA, J&K RERA, JMC, Revenue Dept, SDMA, and OpenStreetMap.

• Task 3: Document initial data acquisition strategy and potential challenges for each source.

• Task 4: Define initial data schema based on abstract variables.

3. Raw Dataset & Mathematical Integrity Findings

An automated, reproducible data was implemented to inspect the supplied dataset (property_dataset_raw.csv). Key quantitative findings include:

• **Dataset Dimensions:** The dataset contains exactly 1,200 rows and 36 columns with zero missing values across all columns.

• **Locality Coverage:** Exactly 36 unique localities are represented, matching the canonical urban and suburban agglomeration of Jammu District.

• **Temporal Reference Year:** Analysis of (year_built + property_age_years) yields an exact constant sum of 2026 across all records, establishing 2026 as the empirical temporal reference baseline.

• **Geospatial Absence:** Property-level geographic coordinates (latitude, longitude) are ABSENT in the raw CSV. Only relative distances to city center and highway are present. Locality-level centroid geocoding is required.

4. Public Data Source Investigation & Provenance Register

Each of the seven public data sources identified in the project synopsis was investigated through network probes and licensing audits:

Source ID & Name	Authority / Portal	Format	Access & Governance Assessment
SRC-01: JDA Approvals	Jammu Dev Authority	HTML / Gazette	Official colony layout schemes and Master Plan 2032 zones verified on jda.jk.gov.in.
SRC-02: J&K RERA	J&K RERA HUDD	Web / PDF	Public J&K RERA project and promoter compliance records.
SRC-03: OpenStreetMap	OSM Foundation	GeoJSON / Vector	Overpass API endpoint operational; complies with ODbL attribution requirements.

https://www.arctechnolabs.com/99acres-real-estate-datasets-scraping.php

https://www.arctechnolabs.com/zillow-real-estate-datasets-scraping.php

https://www.arctechnolabs.com/realtor-real-estate-datasets-scraping.php

5. Problems Encountered, Root Causes & Resolutions

Problem: Limited Access to Real-World Real-Estate Datasets.

Root Cause: Real-estate datasets from platforms such as 99acres, Zillow, and Realtor, accessed through third-party data providers, require paid subscriptions or commercial licensing. The associated costs exceed the available project budget, restricting access to real-world property listings for model development and evaluation.

Resolution: The project will use dummy/synthetic property data for initial development, testing, and validation of the property buy-worthiness assessment and property value forecasting modules. The synthetic dataset will support the implementation of data preprocessing, feature engineering, model development, and system integration. Publicly accessible government and civic data sources will be explored where feasible. The use of synthetic data will be documented as a project limitation, as model performance on dummy data may not accurately reflect real-world market conditions.

6. Conclusion & Verification Checklist

Week 1 setup and data source identification tasks have been executed. All artifacts exist at stated paths in the repository.

- Git repository and modular project structure initialized.
- Raw dataset mathematically audited across 1,200 rows and 36 localities.